



INTRODUCTION TO DATA SCIENCE FINAL PROJECT

Using Data Science to Improve Road Safety: A Case Study on U.S. Car Accidents

BY

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1. Introduction

Car accidents remain a leading concern in the United States, affecting thousands of lives every year and having substantial economic costs. Variability in the complexity of these accidents may depend on many factors, including weather conditions, time of day, traffic patterns, and road design. The research is based on the most extensive dataset available on Kaggle, containing more than 500,000 accident records with 46 features regarding weather conditions, traffic calming measures, accident severity, and geographical locations to pinpoint vital risk factors and trends.

The main aim is to understand how weather, time, and road conditions contribute to the severity of accidents and classify them into risk categories. This study will help city planners and policymakers take strategic measures to improve road infrastructure, optimize traffic control measures, and raise public awareness to improve road safety and reduce the frequency of high-severity accidents through detailed visualizations and data-driven analysis.

2. Objectives

- Identify accident patterns across various dimensions such as severity, weather conditions, and time.
- Explore geographical distributions to pinpoint accident hotspots.
- Provide actionable recommendations to improve road safety based on insights derived from the data
- Developing machine learning models to classify accidents based on severity levels, providing insights into high, medium, and low-risk conditions.
- Investigate time-based patterns to determine peak accident hours and assess variations across weekdays and weekends.
- Create clear and impactful visualizations to communicate key findings and patterns effectively

3. Papers Overview

PAPER 1: "A study on road accident prediction and contributing factors using explainable machine learning models: analysis and performance"

Overview: This paper investigates the use of ensemble machine learning (ML) models to predict road accident injury severity using data from the New Zealand (NZ) road accident dataset. The study aims to predict accident severity based on various contributing factors such as weather, time of day, road type, and number of vehicles involved. The researchers explore a range of ML models, including Random Forest (RF), and Decision Tree evaluating their prediction accuracy and sensitivity. The paper also applies Explainable Machine Learning (XML) techniques, particularly Shapley Value Analysis, to interpret the importance and interaction of features contributing to road accident severity.

Applications:

- **Road Safety Improvements:** The findings can be used to enhance traffic safety measures by identifying key factors influencing accident severity.
- **Traffic Planning:** Insights can help policymakers make informed decisions regarding road design, traffic control, and infrastructure investment.
- **Predictive Analytics:** The study can be applied in real-time accident prediction systems to assist emergency responders and traffic management.

Findings:

- **Best Performing Model:** Random Forest (RF) outperformed other models with 81.45% accuracy, 81.68% precision, 81.42% recall, and 81.04% F1-Score.
- **Feature Importance:** Road category and the number of vehicles involved were found to be significant factors affecting injury severity.

PAPER 2: Modeling Road Accident Severity with Comparisons of Logistic Regression, Decision Tree and Random Forest

Overview: This study explores the prediction of road accident severity using three machine learning models: Decision Tree (DT) and Random Forest (RF). The main focus is on comparing the prediction capabilities of these models, particularly in terms of their ability to handle significant variables. The research aims to identify the most influential variables

for predicting road accident severity and evaluate how the choice of variables affects the performance of each model.

Applications:

1. **Road Accident Severity Prediction:** The models are applied to predict the severity of road accidents based on various factors.
2. **Model Comparison:** It compares the effectiveness of statistical (DT) and nonparametric (RF) methods in predicting outcomes.
3. **Variable Selection:** The paper evaluates the impact of selecting significant variables on the model's prediction performance.

Findings:

1. **Random Forest (RF) Performance:** RF consistently outperforms DT in terms of accuracy, sensitivity, and specificity. It achieves the highest accuracy (74.43%) when using nine significant variables identified by Decision Tree.

4. METHODOLOGY:

- **Data Collection:** Use the US_Accidents_March23_sampled dataset for predicting the severity and factors involved in car accidents.
- **Data Preprocessing:** Clean the data, handle missing values, encode categorical features, and scale numerical values for consistency across models.
- **Model Selection:** Use classification models such as Decision Trees and Random Forests to predict accident severity.
- **Model Evaluation:** Evaluate models using accuracy, precision, recall, and F1-score to determine the best-performing model.
- **Result Interpretation:** Analyze and visualize the impact of key features like weather conditions, time of day, and road type on accident severity prediction.

5. IMPLEMENTATION DETAILS

Dataset Overview:

Size: 500,000 records, 46 columns.

Features:

- **Numerical:** Distance (mi), Temperature (°F), Pressure (in).
- **Categorical:** Severity, City, Weather Condition.

- Binary: Amenity, Crossing, Junction.
- Target Variable: Severity (1: least severe, 4: most severe).

```
[47]: # Quick overview
df = data
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 46 columns):
#   Column                               Non-Null Count  Dtype
---  -
0   ID                                   500000 non-null object
1   Source                              500000 non-null object
2   Severity                             500000 non-null int64
3   Start_Time                          500000 non-null object
4   End_Time                            500000 non-null object
5   Start_Lat                           500000 non-null float64
6   Start_Lng                           500000 non-null float64
7   End_Lat                             279623 non-null float64
8   End_Lng                             279623 non-null float64
9   Distance(mi)                        500000 non-null float64
10  Description                          499999 non-null object
11  Street                              499309 non-null object
12  City                                499981 non-null object
13  County                              500000 non-null object
14  State                               500000 non-null object
15  Zipcode                             499884 non-null object
16  Country                             500000 non-null object
17  Timezone                            499493 non-null object
18  Airport_Code                        498554 non-null object
19  Weather_Timestamp                   492326 non-null object
20  Temperature(F)                      489534 non-null float64
21  Wind_Chill(F)                       370983 non-null float64
22  Humidity(%)                         488870 non-null float64
23  Pressure(in)                        491072 non-null float64
24  Visibility(mi)                      488709 non-null float64
25  Wind_Direction                      488803 non-null object
26  Wind_Speed(mph)                     463013 non-null float64
27  Precipitation(in)                   357384 non-null float64
28  Weather_Condition                   488899 non-null object
29  Amenity                             500000 non-null bool
30  Bump                                 500000 non-null bool
31  Crossing                             500000 non-null bool
32  Give_Way                            500000 non-null bool
33  Junction                             500000 non-null bool
34  No_Exit                             500000 non-null bool
35  Railway                             500000 non-null bool
36  Roundabout                          500000 non-null bool
37  Station                             500000 non-null bool
38  Stop                                 500000 non-null bool
39  Traffic_Calming                     500000 non-null bool
40  Traffic_Signal                      500000 non-null bool
41  Turning_Loop                        500000 non-null bool
42  Sunrise_Sunset                      498517 non-null object
43  Civil_Twilight                      498517 non-null object
44  Nautical_Twilight                   498517 non-null object
45  Astronomical_Twilight               498517 non-null object
dtypes: bool(13), float64(12), int64(1), object(20)
memory usage: 132.1+ MB
None
```

Data Preprocessing:

1. Handling Missing Values:

- Columns with more than 50% missing data were dropped to ensure robust analysis. For other columns with fewer missing values, imputation techniques were applied. For numerical features, the mean was used for imputation and for categorical features, the mode was applied.

```
# Impute numerical columns with median
numerical_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in numerical_cols:
    if df[col].isnull().sum() > 0:
        df[col].fillna(df[col].median(), inplace=True)
```

```
# Impute categorical columns with mode
categorical_cols = df.select_dtypes(include=['object']).columns
for col in categorical_cols:
    if df[col].isnull().sum() > 0:
        df[col].fillna(df[col].mode()[0], inplace=True)
```

2. Categorical Features:

- Categorical variables were encoded using one-hot encoding for features with multiple categories. Binary variables were retained as is. This step ensured compatibility with machine learning models while preserving the interpretability of categorical data.

```
# Define the preprocessing steps for numerical columns (scaling)
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='mean')), # Fill missing values
    ('scaler', StandardScaler()) # Scale numerical features
])

# Define the preprocessing steps for categorical columns (encoding)
categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')), # Fill missing categorical values
    ('onehot', OneHotEncoder(handle_unknown='ignore')) # One-hot encode categorical features
])

preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_cols),
        ('cat', categorical_transformer, categorical_cols)
    ]
)
```

3. Data Splitting:

- The dataset was split into training and testing sets in an 80:20 ratio. This division was done randomly to ensure both sets were representative of the data's distribution.
- Feature scaling was applied to numerical features to ensure consistent scales, especially for tree-based models.

```
# Split the data into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Apply the transformations to the training and test sets
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)

# X_train_processed and X_test_processed are now ready for use in machine learning models
print(f"Processed training data shape: {X_train_processed.shape}")
print(f"Processed test data shape: {X_test_processed.shape}")
```

```
Processed training data shape: (361469, 696862)
Processed test data shape: (90368, 696862)
```

Model Development and Accuracy

1. Decision Tree Classifier:

- A Decision Tree Classifier was developed to predict accident severity. Hyperparameter tuning was conducted to optimize `max_depth` and `min_samples_split`.
- **Accuracy:** 81.82%

```
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train_subsample, y_train_subsample)

# Predict with Decision Tree
dt_y_pred = dt_model.predict(X_test_processed)

# Evaluate Decision Tree
print("Decision Tree Classifier Results")
print(f"Accuracy: {accuracy_score(y_test, dt_y_pred)}")
```

Decision Tree Classifier Results
Accuracy: 0.8181989199716714

2. Random Forest Classifier:

- A Random Forest Classifier was implemented, leveraging multiple decision trees to improve prediction accuracy. The `n_estimators` and `max_features` parameters were tuned for optimal performance.
- **Accuracy:** 77.69%

```
# Simplified Random Forest
rf_model = RandomForestClassifier(
    n_estimators=100, # Fewer trees
    max_depth=30,    # Limit depth
    max_features="sqrt", # Reduce features per split
    n_jobs=-1,       # Parallel processing
    random_state=42
)

# Train on subsampled data
rf_model.fit(X_train_subsample, y_train_subsample)

# Predict on test data
rf_y_pred = rf_model.predict(X_test_processed)

# Evaluate Random Forest
print("Random Forest Classifier Results")
print(f"Accuracy: {accuracy_score(y_test, rf_y_pred)}")
```

Random Forest Classifier Results
Accuracy: 0.7768900495750708

The **Decision Tree** model achieved an accuracy of **81.8%**, while the **Random Forest** model had an accuracy of **77.7%**. The Decision Tree is simpler, faster, and easier to interpret but is more prone to overfitting, especially with deeper trees.

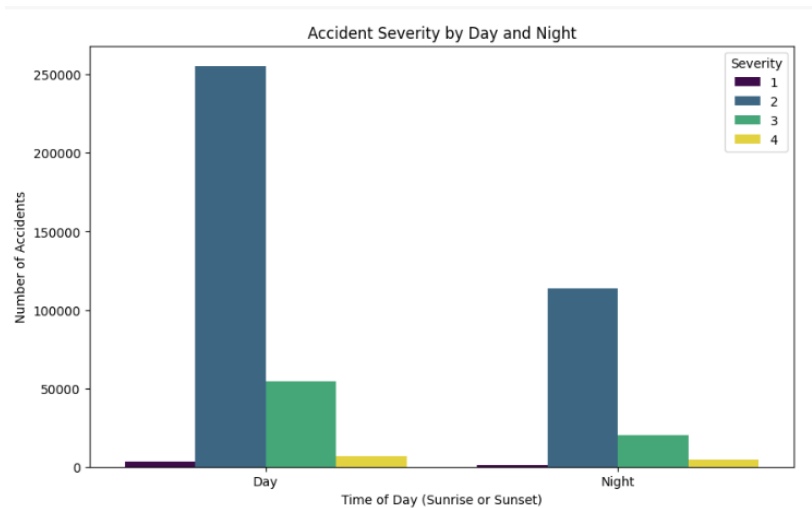
Visualization Techniques

Heatmaps, bar plots, and line charts were utilized to uncover patterns in accident data.

6. Key Findings (from the dataset)

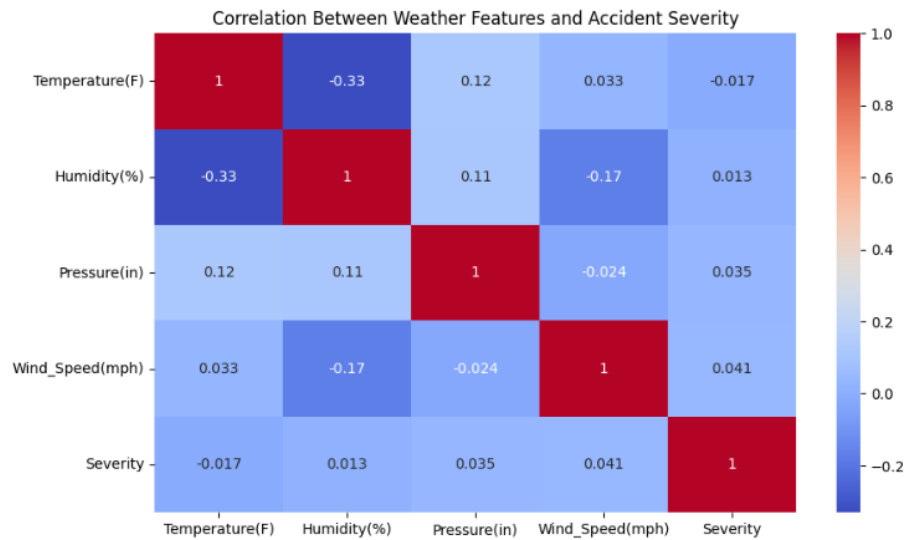
6.1 Accident Severity Analysis

- Severity 2 was the dominant category during both day and night. Most accidents occurred during the day due to higher traffic volumes.



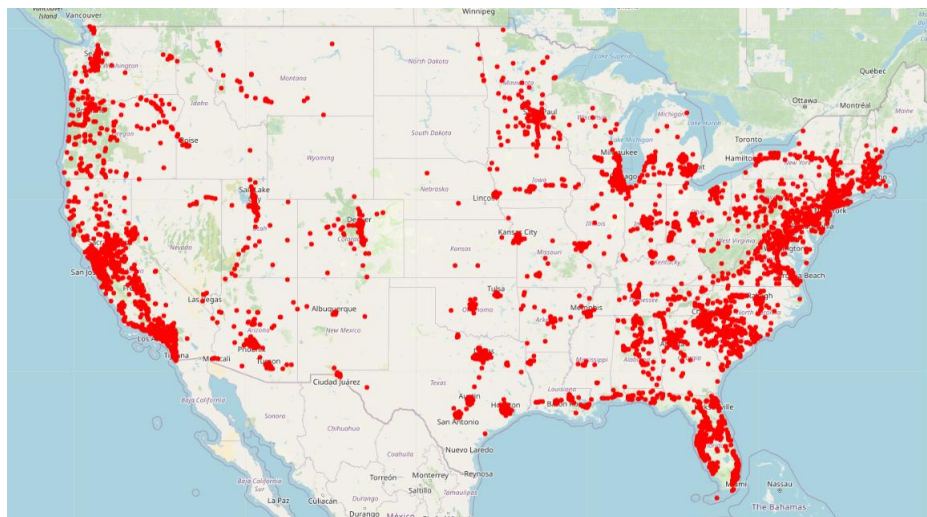
6.2 Correlation Analysis

- Weather features such as temperature and humidity showed minimal correlation with accident severity. This indicates that even though we assumed weather conditions have a big role to play in car accidents, they don't contribute much.
- Wind speed had the highest correlation (0.041) with severity, which is still very weak.



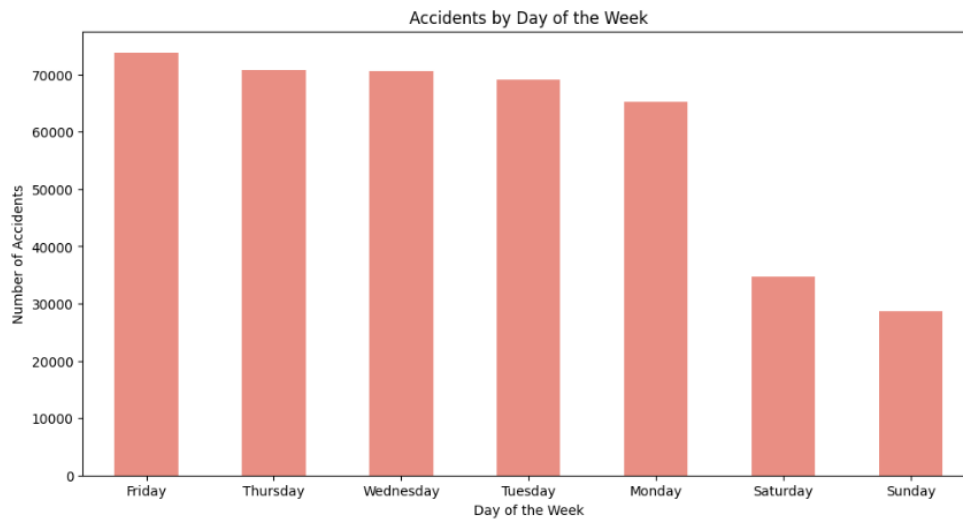
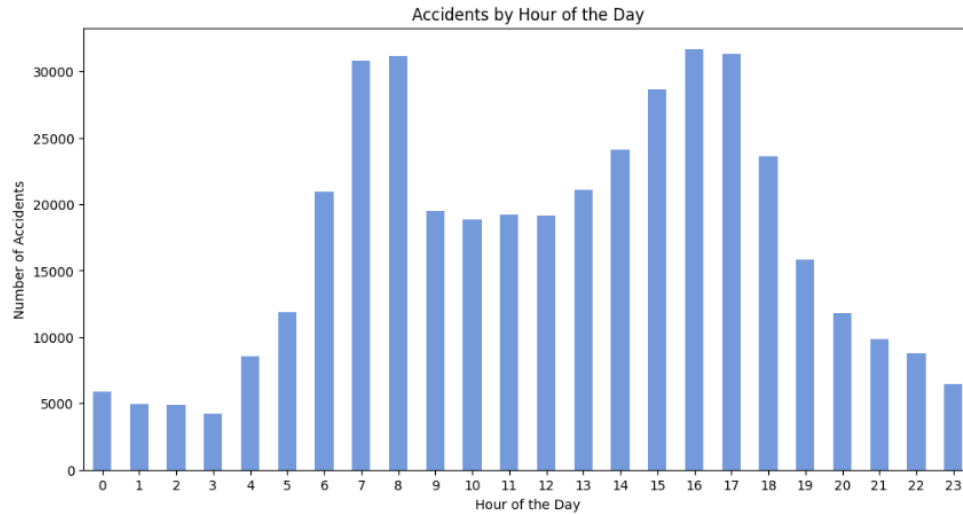
6.3 Geographical Insights

- Cities like Los Angeles, New York City, Miami, and Chicago showed high accident rates.
- The East Coast exhibited denser accident patterns than the Western regions.
- Linear patterns of red marks on maps corresponded to major highways. This means there might be a higher chance that major highways end up having frequent accidents as the cars are always at high speed. Any kind of carelessness or driving style can cause accidents.
- **Note:** The main dataset was too big to work with, so we randomly took 1,00,000 rows to draw this map.



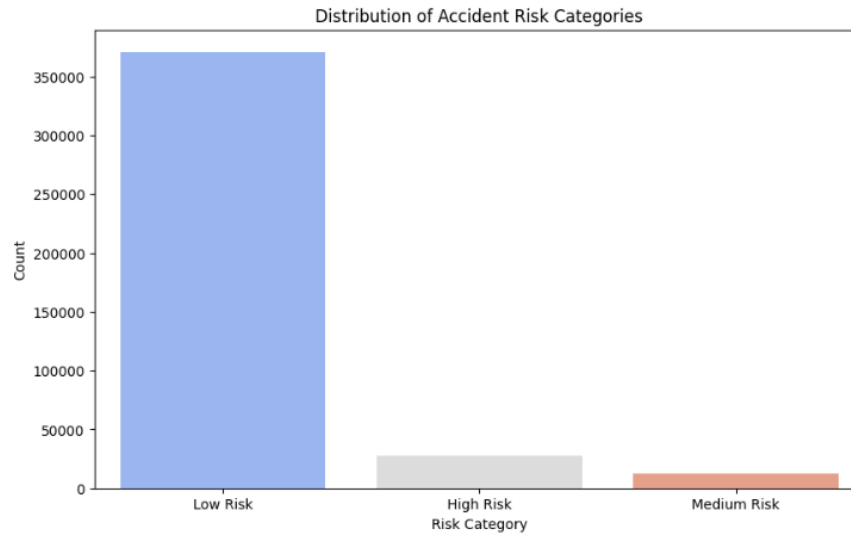
6.4 Time-Based Patterns

- Peak accident hours were 7-8 AM and 4-5 PM, correlating with heavy commuting times.
- Significant reduction in accidents at weekends, with Fridays having the highest reports.



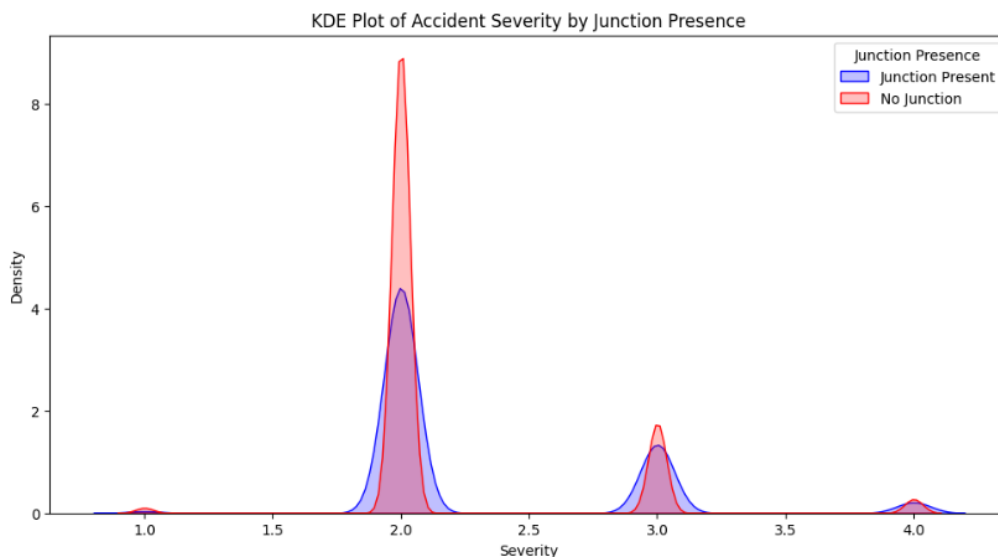
6.5 Accident Risk Categories

- Most accidents occurred under low-risk conditions.
- Severe accidents were fewer but had more severe outcomes.



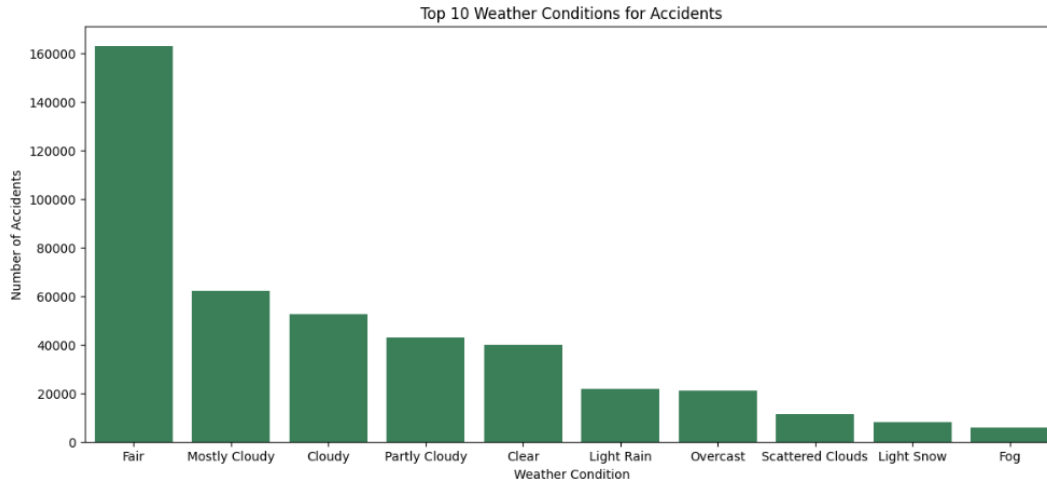
6.6 Junction Presence Analysis

- Both "Junction Present" and "No Junction" cases show the highest density at Severity 2, indicating moderately severe accidents.
- Accidents occur more frequently in areas without junctions



6.7 Weather Condition Analysis

- Most accidents occurred under "Fair" weather conditions, followed by "Cloudy" and "Rain." High-risk weather conditions like snow or storms were less frequent & people need to be more focused while driving.



7. Challenges:

This analysis had some challenges, like missing data on important features, which required careful handling to avoid errors. The dataset's large size of 500,000 records made processing and visualization more complex. Weak links between weather and severity also showed that other factors needed to be analyzed. Despite these issues, useful patterns and insights were discovered.

8. Recommendations

- Introducing more safety measures like traffic calming in high-risk areas during daytime.
- Enhancing traffic monitoring and safety measures on highways.
- Implement targeted road safety measures during peak commuting hours on working days.
- City planners could introduce more roundabouts or similar traffic control measures to reduce vehicle speed and prevent collisions.
- Adding roundabouts or signal-controlled junctions to accident-prone areas.
- Enhancing public awareness and emergency planning during adverse weather conditions.

9. References:

<https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents/data>

<https://www.kaggle.com/code/aymanragab/california-traffic-accident-analysis>

<https://www.kaggle.com/code/mariusborel/us-accident-state-and-year-maps>